Developing an AI Agent to Support Administration and Operation of HPC Systems Based on SLURM

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Motivation

- Slurm CLI is complex — dozens of commands, flags, Bash scripting
- No safety layer for destructive operations (cancel, hold, release)
- Open-weight LLMs lack Slurm tool-calling knowledge
- No standardised benchmark for NL-based Slurm agents

Approach

- Observer/Operator Split: Read-only Observer + state-changing Operator; destructive actions require human confirmation
- Grounded RAG: 273 official Slurm docs → 2,276 chunks; reduces hallucination
- Fine-Tuning: Qwen2.5-14B-Instruct + QLoRA from GPT-5-mini teacher traces
- Benchmark: 3,135 cases, 11 categories, LLM-as-judge scoring

Implementation

- Stack: React/Vite → FastAPI → MCP → real Slurm cluster
- Tools: 64 agent-facing tools (29 read-only for Observer, 35 action + 5 discovery reads for Operator)
- Training: 2,625 unique samples · 3 epochs · A40 48 GB · QLoRA 4-bit NF4 · r=64 all-layer · 8,192 ctx · ~$20

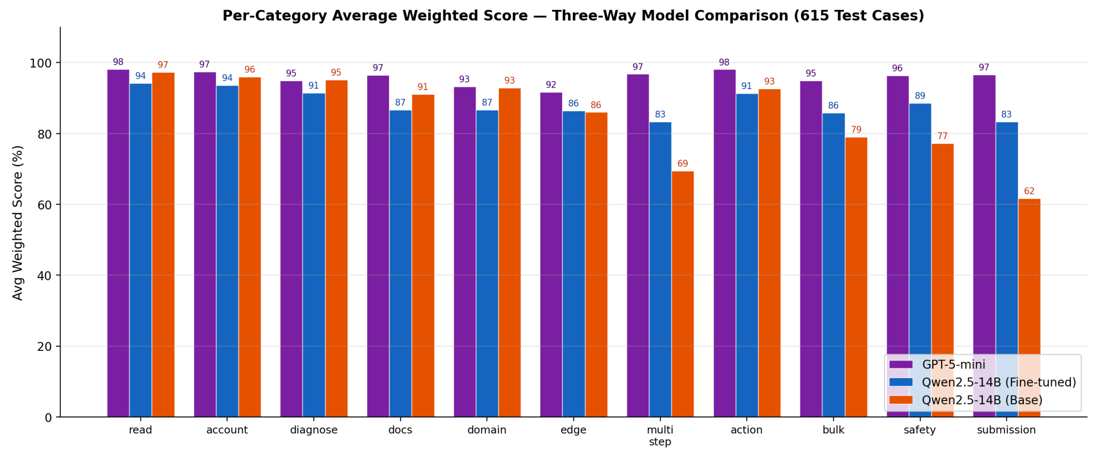
Results — Three-Way Model Comparison

615-case held-out test set (stratified 80/20 split, seed=42)

Model	Score	Tool	HITL	State	Judge	Lat.
GPT-5-mini	95.9%	98.8%	99.0%	98.2%	75.7%	28s
FT Qwen (ours)	88.3%	89.2%	90.4%	90.0%	79.7%	85s
Base Qwen	85.2%	84.9%	90.2%	89.8%	76.8%	81s
Monolithic ablation‡	81.6%	77.5%	88.9%	78.9%	59.2%	77s

‡ Routing excluded; re-normalised weights.

- FT closes 29% of gap to GPT-5-mini
- O/O beats mono by +6.1 pp; +10.2 pp routing-neutral
- FT judge (79.7%) exceeds base (76.8%) & commercial (75.7%)
- All pairwise differences significant: Wilcoxon $p \ll 0.001$, paired t-test $p \ll 0.001$, Bonferroni-corrected



Three-Way Per-Category Comparison

Contributions

- Novel Observer/Operator dual-agent safety architecture for Slurm
- HITL confirmation workflow for all destructive operations
- Grounded RAG pipeline over 273 official Slurm docs
- 3,135-case curated benchmark with LLM-as-judge scoring
- Cost-effective fine-tuning (~$20) achieving 88.3% mean weighted score

Future Works

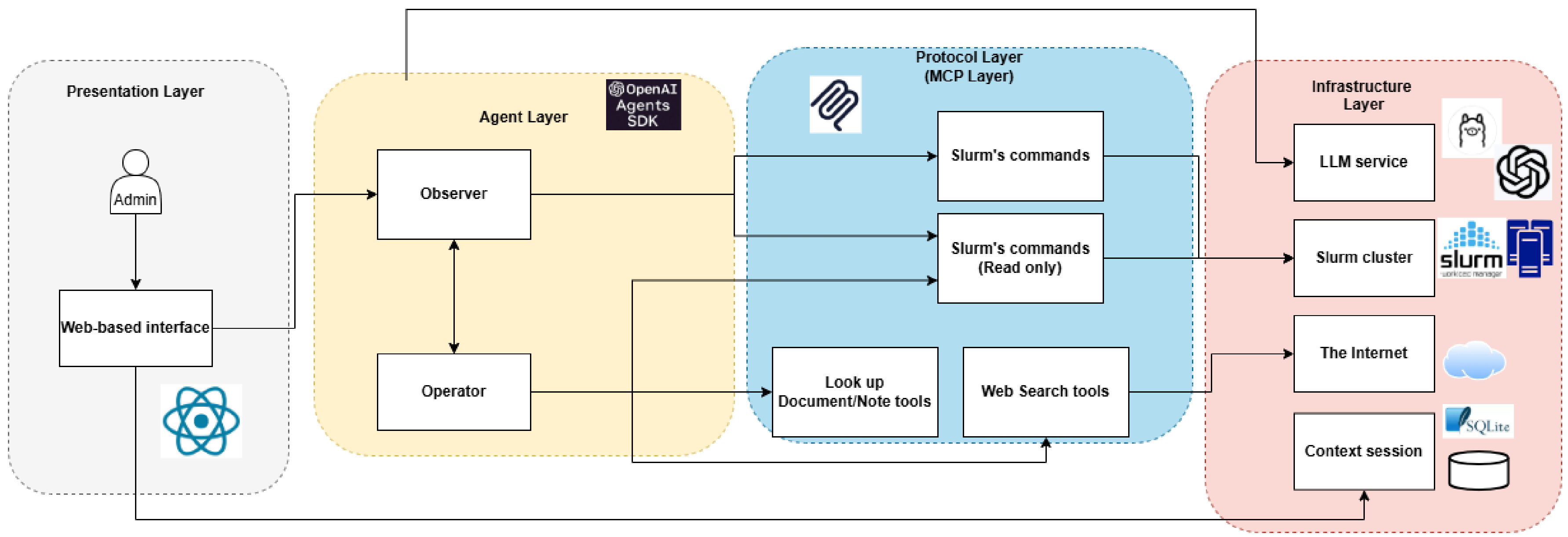
- Multi-tenant authentication, resource isolation, audit logging
- Practitioner user studies for usability validation
- Expand benchmark to cover more Slurm edge cases
- Explore larger open-weight models and multi-turn fine-tuning

Scoring Formula

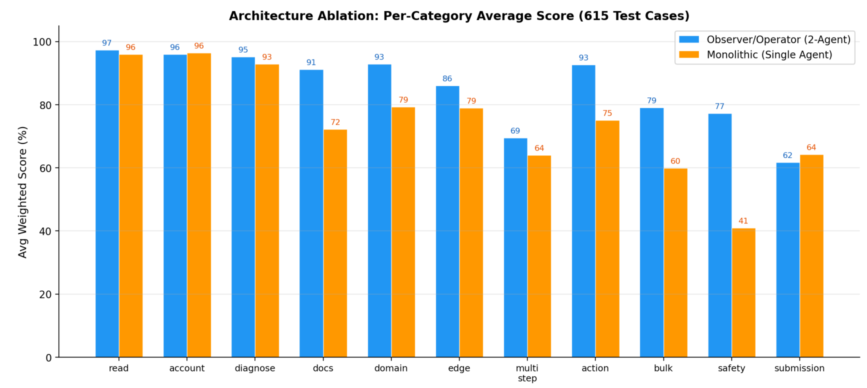
$$s = 0.35\, r_{\text{tool}} + 0.25\, r_{\text{route}} + 0.25\, r_{\text{HITL}} + 0.15\, r_{\text{state}}$$

Ablation (mono): routing removed, weights $\times 4/3$

$$s_{\text{abl}} = \tfrac{7}{15}\, r_{\text{tool}} + \tfrac{1}{3}\, r_{\text{HITL}} + \tfrac{1}{5}\, r_{\text{state}}$$



System Architecture



Ablation: Dual-Agent vs. Monolithic